

Spatial and Temporal Distribution of Forest Fire in Terai and Chure Range, Nepal

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Abstract

Fire is the most spectacular natural hazard that deleteriously affects forest ecosystem, species composition and ultimately biodiversity. Hence, mapping of forest fire risk zone is required and thereby minimizing the frequency as well as intensity of forest fire. This study focused on distribution of fire, its frequency and magnitude on Terai and Churia range. For this, Moderate Resolution Imaging Spectro-radiometer (MODIS) on the Terra and Aqua satellite data having specific features for monitoring and recording the details and location of fire hotspots have been used. Vector data of forest fire obtained from MODIS were analysed using Arc GIS and area for each feature according to month and yearly basis were calculated. Magnitude of area burnt by fire was used to find out the high, medium and low fire risk zone map of study area.

The results show that there were altogether 9463 hotspots covering 5,79,845 ha throughout the last 15 years. Within the study period 2005, 2009, 2012 and 2013, forest fires left historically largest annual burnt area i.e. 280,955 ha (48.45%). March, April and May has triggered the highest count 8675 ha (91.67%). Most of the fire incidence was inside protected areas (PAs). Figuratively, total fire hotspot inside and outside PA was 4879 and 4584 respectively. Total forest area covered by fire inside PAs was 326,152 ha while 253,692 ha area was detected outside PAs. Bardai National Park has the highest magnitude of burnt area and fire hotspot in comparison to other PAs of Terai and Churia range. Western Region has highest fire incidence in comparison to Eastern region. Hence special consideration must be taken by the forest department, park authority and other concerned stakeholder during hot and dry season with well-equipped fire crew. Awareness and capacity building programme should also be undertaken to minimize forest fire hazards.

Keywords: MODIS, Spatial, Temporal, Forest fire Hotspot, fire incidence.

Introduction

Fire has fascinated us ever since early humans had the idea of rubbing two sticks together. Since the dawn of human civilization, people are using fire intentionally or unintentionally. Charles Darwin 1882, describes the discovery of fire as "probably the greatest ever made by man, excepting language" The proverb "fire is the good servant but bad master" advocate that, fire under control may be beneficial but uncontrolled fire is always harmful. Fire is therefore, neither destructive nor constructive; it simply causes changes. Whether these changes are viewed as desirable or not depends upon their compatibility with overall objectives (Wade and Lundsford, 1990). Forest fires recur year after year, with a devastating intensity. Despite the fact that during the last decades scarce precipitations along with high temperature levels have been impacting the fire risk (Moriondo *et al.*, 2006), these two variables represent just some of the possible causes of forest fires.

Fire risk depends on a number of regional specific factors (Moreno *et al.*, 1998 and Martinez *et al.*, 2008 and 2009; Westerling *et al.*, 2006 for US, among others), human attitudes (Barbero *et al.*, 1990; Martinez *et al.*,

2008 and 2009; Pausas and Keeley, 2009), and weather patterns (Westerling *et al.*, 2006; Pausas and Fernández-Muñoz, 2012). Nepal has always been vulnerable to several types of natural disasters—including, landslides, hailstorms, floods, drought and fire - and according to projections the quantity and intensity of these events will increase with climate change. (NARC, 2010). The combination of all these aspects is therefore responsible for generating the final scenario of fire danger.

The spatial pattern of forest fire occurrence is a key factor in understanding forest fire dynamics, and presence of a forest fire is determined by several biotic and abiotic factors, however, the effects of each factor vary between ecosystems and within spatial and temporal scales (Yang *et al.*, 2007). There is still a lack of understanding of the fire phenomenon, its impacts and what should be done to address these problems. In addition, fires are responsible for shaping out most of the ecosystem, they can be destructive both to vegetation and human life (Boschetti *et al.*, 2008), Carmel *et al.*, 2009). A fire has a major role in the global carbon cycle (Lu *et al.*, 2006, Merino-de-Miguel *et al.*, 2010) and are a primary disturbance in most forest ecosystems (Fraser and Li, 2002). There is a growing debate on the relationship between fire and

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climate change (Weber and Flannigan, 1997; Siegert *et al.*, 2001; Alencar *et al.*, 2006; Westerling *et al.*, 2006; Denman *et al.*, 2007) and a perceived increasing incidence, extent, and severity of uncontrolled burning globally that has led to calls for international environmental policy concerning fire (FAO, 2007).

Methodology

Study area

Terai and Churia lies in southern part of Nepal (figure 1). Terai and Churia occupies 20,16,998 ha (13.70%) and 18,98,263 ha (12.84 %) of total land area of Nepal respectively. In Terai, forest, other wooded land and other land cover are 4,11,580 ha (20.41%), 9,502 (0.47%) and 15,95,916 ha (79.12%) respectively whereas in Churia, forest, other wooded land and other land cover are 13,73,743 ha (72.37%), 22,672 ha (1.19%) and 5,01,848 ha (26.44%) respectively (DFRS/FRA, 2014).

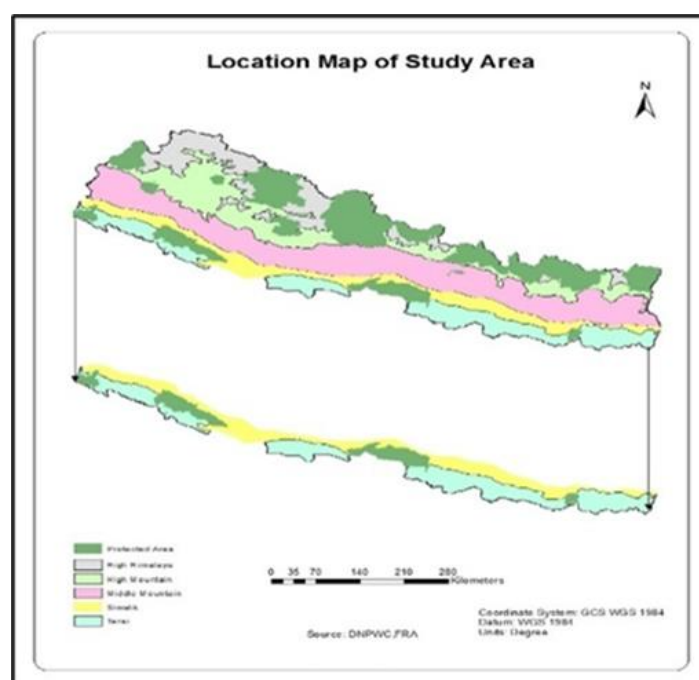


Figure 1. Location map of study area.

Nepal has established a very good network of Protected Areas system with 12 National Parks, 1 Wildlife Reserve, 1 Hunting Reserve, 6 Conservation Areas, and 13 Buffer Zones extending from lowland Terai to high mountains, covering 23.39 % of the total country's land, which contribute to in-situ conservation of ecosystems and biodiversity across the country. Conservation efforts made by the government of Nepal is worldwide popular and highly recognized by the international societies (DNPWC).

Data and Software Used: The NASA Moderate Resolution Imaging Spectroradiometer (MODIS) which is a multiple band radiometric sensor instrumental board Terra (NASA launched on December 18, 1999, at local time 10:30AM) and Aqua (NASA launched on May 4,

2002, at local time 1:30 PM) was used. These Terra (morning) and Aqua (afternoon) satellites have specific features for fire monitoring and have been used systematically to generate a suite of global MODIS land products (Justice *et al.*, 2002b) which was downloaded from official website of NASA. The forest cover data of Terai and Churia was obtained from Department of Forest Research and Survey (DFRS) and map of the Protected Areas (PAs) were obtained from Department of National Park and Wildlife Conservation (DNPWC). Arc-GIS was used for geo-spatial data analysis and output, MS office for analysis, writing and presentation while SPSS and R-software was used for statistical analysis.

Research Methods

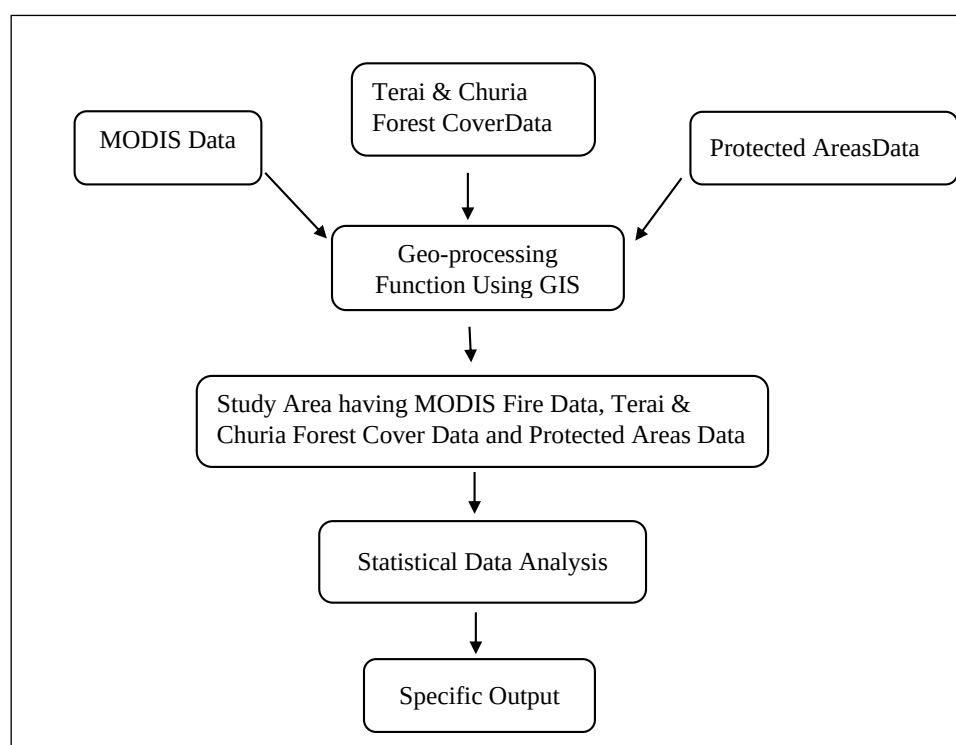


Figure 2. Stepwise methodology followed.

From the official website, whole burnt area of South East Asia was extracted and converted to vector data in GIS format using Arc GIS 10.2.2 with an attribute containing magnitude and distribution of fire. Fire data on the area of interest i.e., Terai and Churia forest cover including PAs were further obtained by Geo-processing function (clip, merge, intersect, union, dissolve) of ARC GIS and the centroid of each burnt area were calculated. Whole data were projected in the WGS 1984 and Degree and all monthly data of 2000 to 2014 were merged into a single layer (figure 2).

The field work consists of ground verification for the sake of accurate ground truthing. Seven Places were randomly selected and the ground verification was done over there in order to check the accuracy of MODIS data.

Statistical analysis was done to obtain the spatial and temporal distribution, magnitude of area

burnt. After getting all the results of magnitude of burnt areas, fire distribution regarding monthly and yearly statistical data of forest fire and fire risk zone map of study area were identified.

Results and Discussion

Magnitude of burnt area in Terai and Churia Range

The magnitude of burnt area and temporal distribution pattern of forest fire from 2000 to 2015 were analysed. The results shows that there were altogether 9463 hotspots covering 5,79,845 ha throughout the last 15 years. The majority of fires (about 65 percent) hotspot falls under Mid-Western Development Region whereas, Eastern Development Region has less fire incidence due to more precipitation (Khawas, 2007), Cumulative frequency of fire distribution pattern for the last 15 year are presented in figure 3.

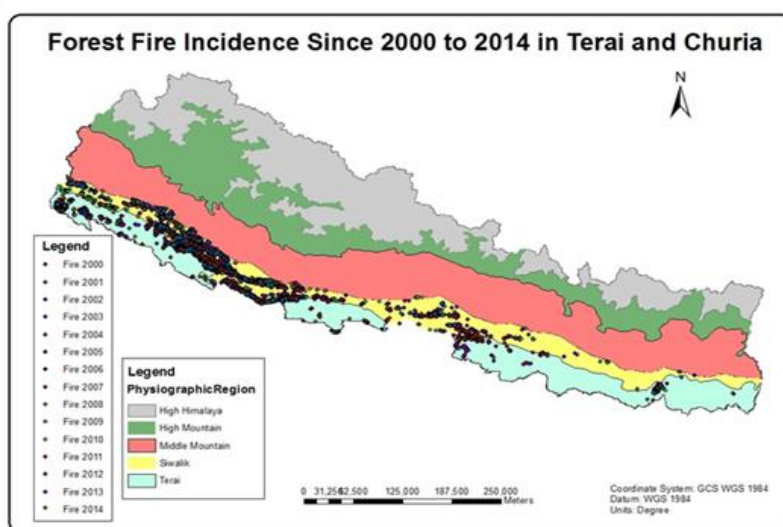


Figure 3. Forest fire incidence since 2000 in Terai and Churia.

The maximum number of forest fire was in 2009 AD which is 1,304 in number and the area burnt was 72,732 Ha. as shown in figure 4.

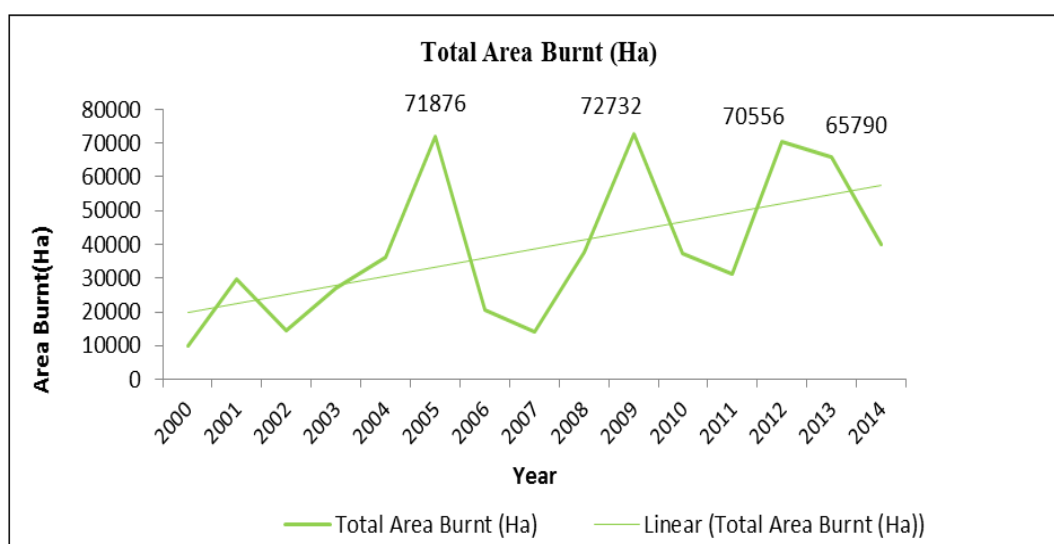


Figure 4. Annual total area burnt.

Fire distribution in Terai and churia region

The total yearly number of fires recorded within Nepal substantially from around 153 at the start of the study period (2000). In 2005, 2009, 2012 and 2013 forest fires left historically largest annual area burnt 280,955 ha, i.e. 48.45 % where March, April and May has triggered the highest fire count 8,675. The annual fire

occurrence during last 15 years (2000-2014) is shown in figure 5.

There was a large inter-annual variation in average monthly patterns of fire occurrence but, the majority of fires were observed during March, April and May. Among them the most significant being April, which had the highest average number of fires.

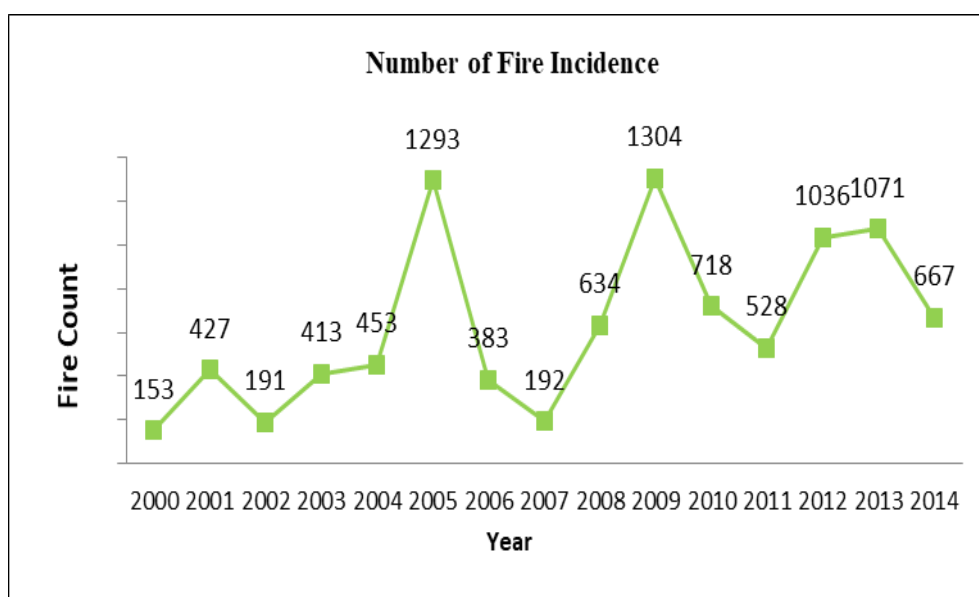


Figure 5. Fire counts in Different Years.

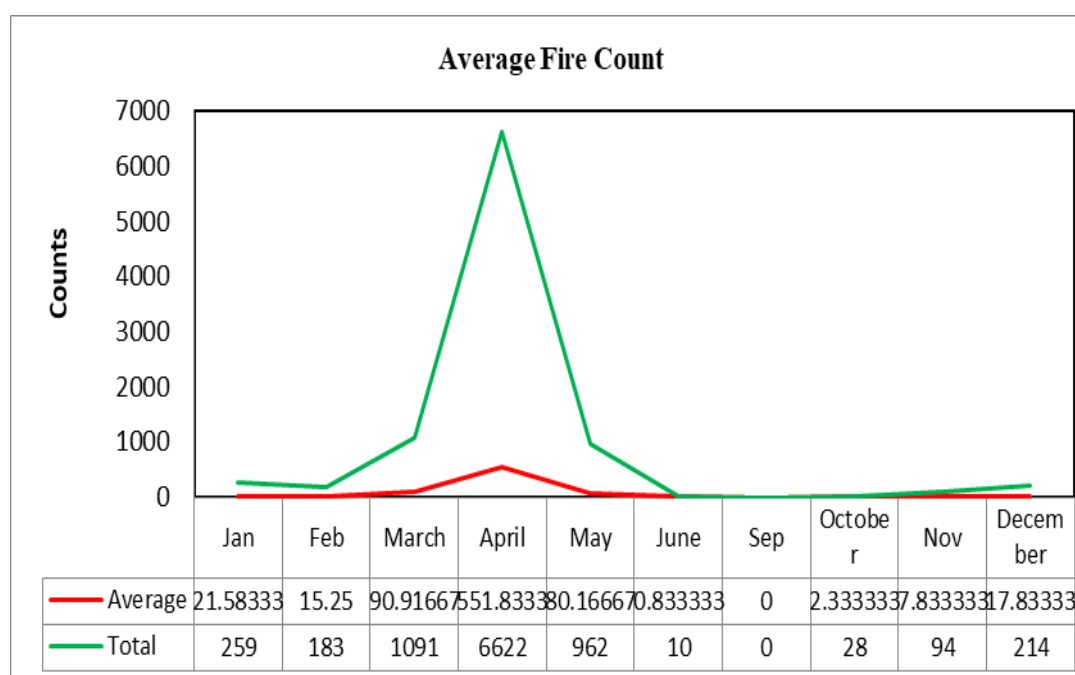


Figure 6. Average Counts of Forest Fire in Different Months.

From figure 6 it is seen that March, April and May has triggered the highest fire count 8,675 which is equivalent to 91.67% of the annual fire incidence. Very few fire incidence were observed in September, October and November

Most of the forest fire were inside forest while non-forest has negligible magnitude of area as shown in figure7.

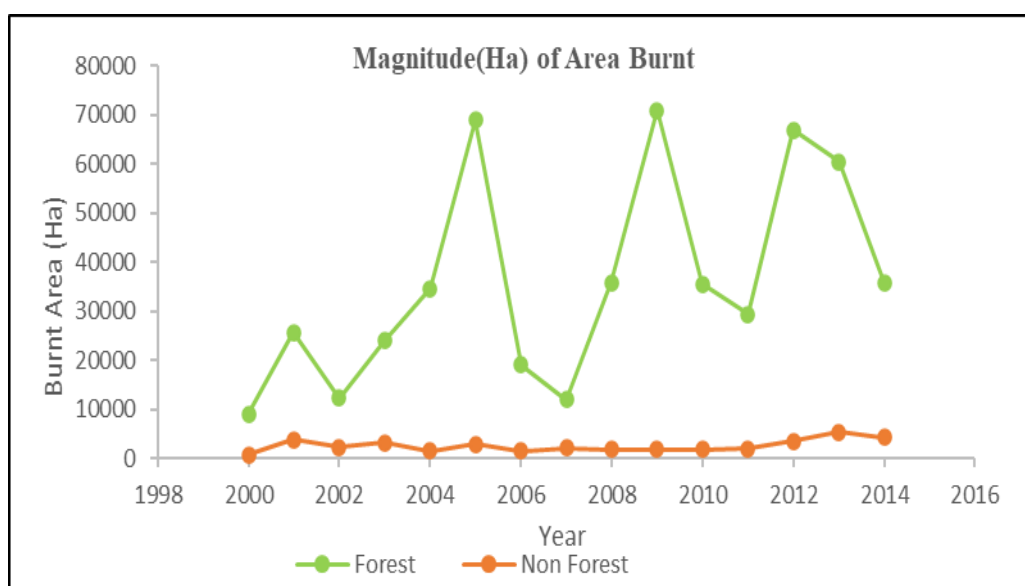


Figure 7. Number of Fire in Forest and Non-Forest Area.

Forest fire in protected area of Terai and Churia range

Distributions pattern of forest fire frequency on PAs were investigated on a monthly and yearly basis from 2000-2014 AD and the monthly changes were studied for each year. Bardia National Park has highest

frequency of fire occurrence while Parsa Wildlife Reserve has lowest fire incidence. In case of Buffer zone, Banke National Park Buffer Zone has highest fire frequency while Koshi Tappu Wildlife Reserve Buffer zone has lowest fire frequency. The fire occurrence and extent of burnt area during the study period inside PAs is shown in figures 8 and 9 respectively.

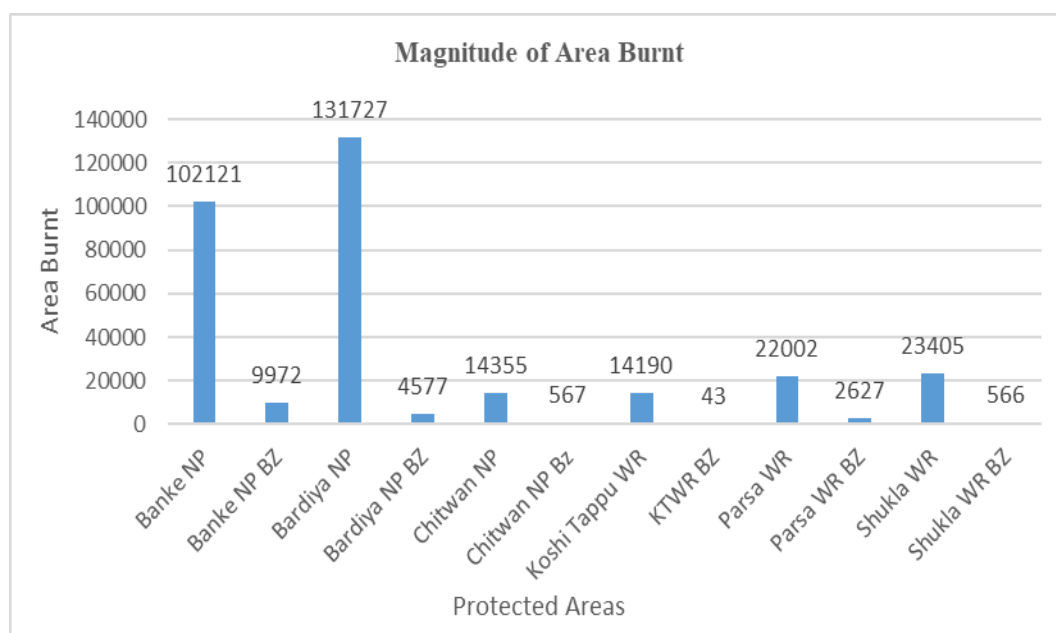


Figure 8. Fire incidence within Protected Areas.

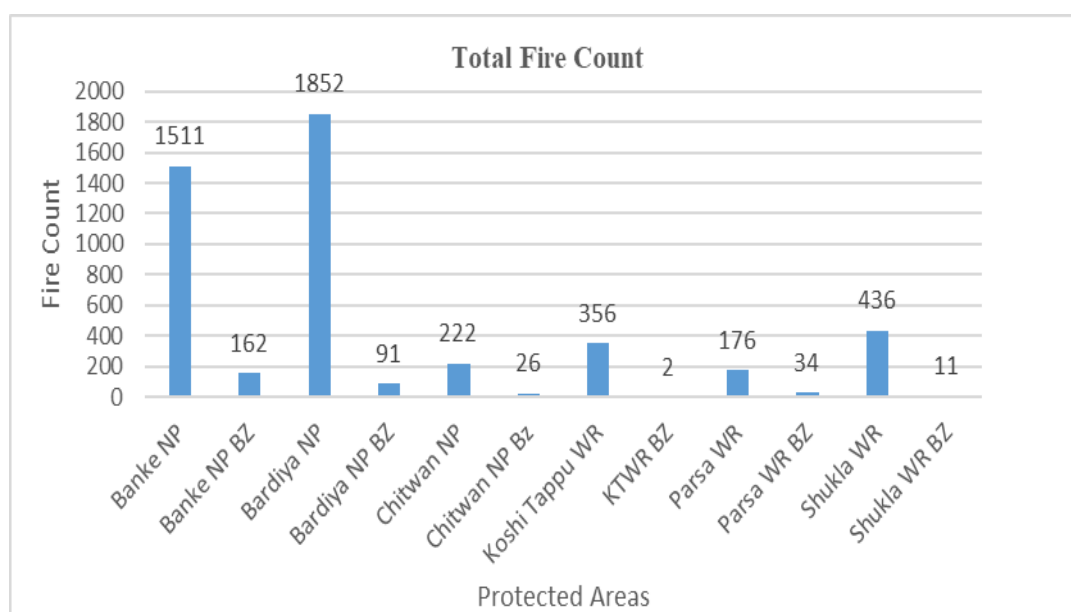


Figure 9. Burnt Area within Protected Areas.

Bardia National Park has the highest magnitude of burnt area in comparison to other protected areas.

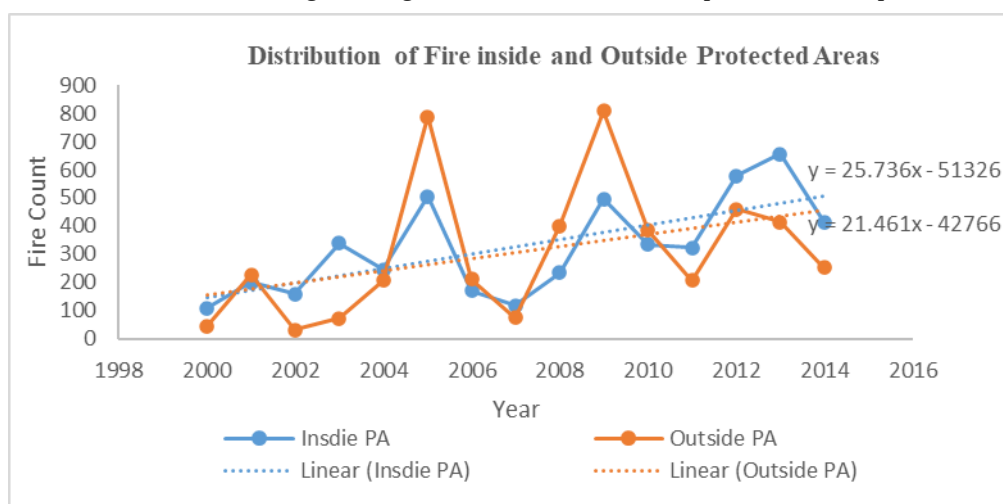


Figure 10. Distribution of Fire inside and outside Protected Areas.

Fire Sensitive Zone in Nepal

The forest risk map in figure11, shows that fire with high-risk zone is in the Mid-western development region while there is low risk zone in Eastern region of

Nepal. Koshi Tappu WLR, Suklaphanta WLR, Banke NP and Bardia NP lies in high sensitive zone whereas, Parsa WLR and Chitwan NP lies in sensitive fire risk zone.

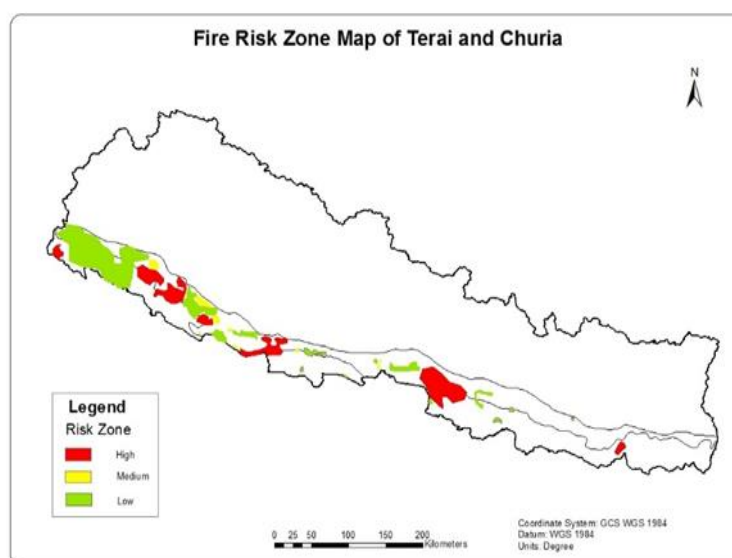


Figure 11. Fire sensitive zones of Nepal.

Conclusions and Recommendations

From the study it can be concluded that there were altogether 9,463 hotspots covering 5,79,845 ha. of forest area throughout the last 15 years. In the year 2005, 2009, 2012 and 2013 forest fires left historically largest annual area burnt i.e. 2,80,955 ha (48.45 %). March, April and May were fire sensitive months and has triggered the highest fire count 8,675 i.e. (91.67%) as this time interval is the most driest period. Most of the fires were inside PA. Total fire count inside and outside PAs was found 4879 covering 3,26,125 ha and 4,584 covering 2,53,692 ha respectively. Bardia NP has highest fire incidence and magnitude of area burnt in comparison to other PAs which lies in Terai and Churia. Mid-Western region of Terai and Churia lies in fire sensitive zone. 91.77% of fires occurred during Hot and Dry season in Terai and Churia. MODIS data and GIS could be successfully employed for fire detection and monitoring fire-prone areas. Hence Special consideration must be taken by the forest department during hot and dry season with well-equipped fire crews. Park authority and concerned stakeholder should focus more on fuel management. Forest department and park authority should develop multi-year program to deal with forest fires. Patrolling should increase more on fire sensitive months, season and areas, Construction of fire lines to suppress the flow of fire from settlements or roads towards forest, within forest and fire prone areas. Government should sanction more budget on modern technologies for the prevention, monitoring, detection and response to forest fire.

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